

The Blessings of Overparameterization: Applications in Solving Economic Models

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Motivation, Question, and Contribution

“I remember my friend Johnny von Neumann used to say, with four parameters I can fit an elephant, and with five I can make him wiggle his trunk.”—*Enrico Fermi*

- Deep learning has proved remarkably capable at solving **high-dimensional problems**.
- Economists are now using it to find accurate approximate solutions to dynamic models once thought intractable.
- These networks are **massively overparameterized**, with far more parameters than “data” points.
- Standard statistical wisdom says: more parameters than “data” points leads to **overfitting** and poor out-of-grid performance.

When solving dynamic models with neural networks

- Does overparameterization lead to poor approximation in out-of-grid performance?
- Is overparameterization a “necessary evil” that comes with modern machine learning methods, or does it give us something extra?

1. **No overfitting:** as the number of network parameters grows, the solutions become more accurate.
2. **Algorithmic stability:** the approximate solutions of dynamic models in economics are unstable; overparameterization increases stability.

Implication

When in doubt, use a bigger network. Overparameterization is not a source of risk to be managed but a feature to be exploited.

Background: Economic Models, Approximate Solutions, and Overparameterization

An economic model is a system of functional equations

Many theoretical models can be written as functional equations:

- Economic object of interest: f , where $f : \mathcal{X} \rightarrow \mathcal{R} \subseteq \mathbb{R}^N$
 - e.g., production, investment choice, best-response, etc.
- Domain of f : $\mathcal{X}$
 - e.g. space of capital, opponents state, etc.
- The “Economics model”: $\ell(x, f)$
 - e.g., Euler and Bellman residuals, equilibrium FOCs.

Then a **solution** is $f^* \in \mathcal{F}$ where $\ell(x, f^*) = \mathbf{0}$ for all $x \in \mathcal{X}$.

Approximate solution

1. Sample $\mathcal{X}$: $\mathcal{D} = \{x_1, \dots, x_N\}$
2. Pick a family of parametric functions (e.g., polynomials, Chebyshev polynomials, neural networks) $f_\theta(\cdot)$:
 - $\theta = \{\theta_1, \dots, \theta_M\}$: parameters for optimization (e.g., coefficients, or weights and biases).
3. To find an approximation for f solve:

$$\min_{\theta} \frac{1}{N} \sum_{x \in \mathcal{D}} \left\| \underbrace{\ell(x, f_\theta)}_{\text{Econ model error}} \right\|_2^2$$

Neural Networks and Overparameterization

Neural Networks form a **highly-overparameterized** ($M \gg N$) class of functions.

- Example: one layer neural network, $f_\theta : \mathbb{R}^Q \rightarrow \mathbb{R}$:

$$f_\theta(x) = W_2 \cdot \sigma(W_1 \cdot x + b_1) + b_2$$

- $W_1 \in \mathbb{R}^{P \times Q}$, $b_1 \in \mathbb{R}^{P \times 1}$, $W_2 \in \mathbb{R}^{1 \times P}$, and $b_2 \in \mathbb{R}$.
- $\theta \equiv \{b_1, W_1, b_2, W_2\}$ are the coefficients, in this example $M = PQ + P + P + 1$.
- $\sigma(\cdot)$ is a nonlinear function applied element-wise (e.g., $\max\{\cdot, 0\}$).
- Making it “deeper” by adding another “layer”: $f_\theta(x) \equiv W_3 \cdot \sigma(W_2 \cdot \sigma(W_1 \cdot x + b_1) + b_2) + b_3$.

Applications

- We demonstrate these properties across three canonical models with known benchmarks:

1. Linear-Quadratic

Closed-form solution. Sharpest test of approximation error.

2. McCall Job Search

1D state, kink at reservation wage. Non-smooth approximation.

3. Real Business Cycle

2D stochastic state. No closed form. Benchmarked vs. VFI.

- In all three: **accuracy** and **stability** improve with overparameterization.

Application 1: Linear-Quadratic regulator

The LQ problem

- The general deterministic LQ problem:

$$\begin{aligned} v(x) &= \max_u \{ -x^\top R x - u^\top Q u + \beta v(x') \} \\ \text{s.t. } x' &= Ax + Bu, \quad x_0 \text{ given} \end{aligned}$$

- State variable: $x \in \mathbb{R}^n$. Control variable: $u \in \mathbb{R}^k$.
- Closed-form solution:

$$v(x) = -x^\top P x, \quad u(x) = -F x$$

where P and F are obtained from the discrete-time algebraic Riccati equation.

- Sharpest test: approximation error can be computed exactly at every point.

A competitive firm with adjustment costs

- A price-taking firm in a competitive industry.
- Own output y , aggregate output Y (exogenous). Inverse demand: $\alpha_0 - \alpha_1 Y$.
- The firm chooses investment u subject to quadratic adjustment cost $\frac{\gamma}{2}u^2$:

$$\begin{aligned} v(y, Y) = \max_u & \left\{ (\alpha_0 - \alpha_1 Y)y - \frac{\gamma}{2}u^2 + \beta v(y', Y') \right\} \\ \text{s.t.} \quad & Y' = h_0 + h_1 Y, \quad y' = y + u \end{aligned}$$

- State is two-dimensional: (y, Y) . With augmented state $x = (1, y, Y)^\top$, this maps into the general LQ framework.
- Euler equation:

$$\gamma u(y, Y) = \beta [\gamma u(y', Y') + (\alpha_0 - \alpha_1 Y')]$$

LQ: neural network implementation

- **Architecture:** one-hidden-layer ReLU networks for value and policy functions. Hidden units $H = 1$ to $H = 7$ (4 to 22 parameters).
- **Training data:** the optimal u depends only on Y (y enters linearly and does not affect the marginal return to investment), so we minimize the mean squared Euler residual over $n_Y = 6$ grid points:

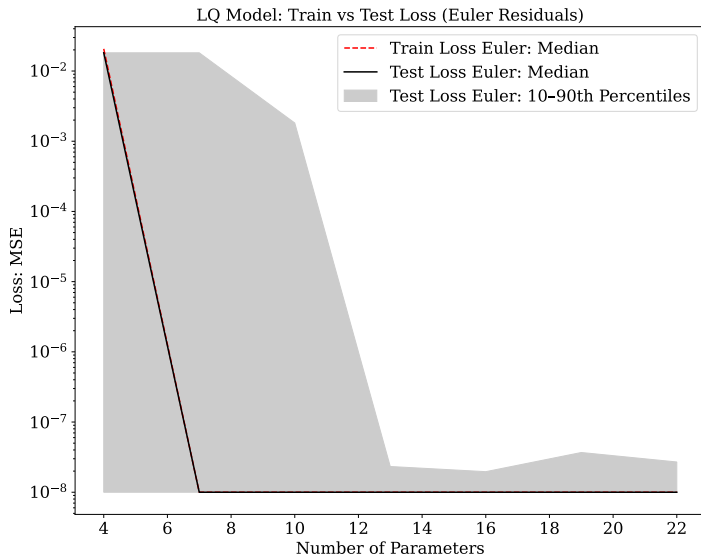
$$\min_{\theta_u} \frac{1}{n_Y} \sum_{i=1}^{n_Y} [\gamma u(Y_i; \theta_u) - \beta(\gamma u(Y'_i; \theta_u) + \alpha_0 - \alpha_1 Y'_i)]^2$$

- Adam optimizer with early stopping at loss $< 10^{-8}$.
- **Test data:** $T = 29$ period path simulated from fixed initial conditions. Test points lie in the interior of the state space and never on a training node.

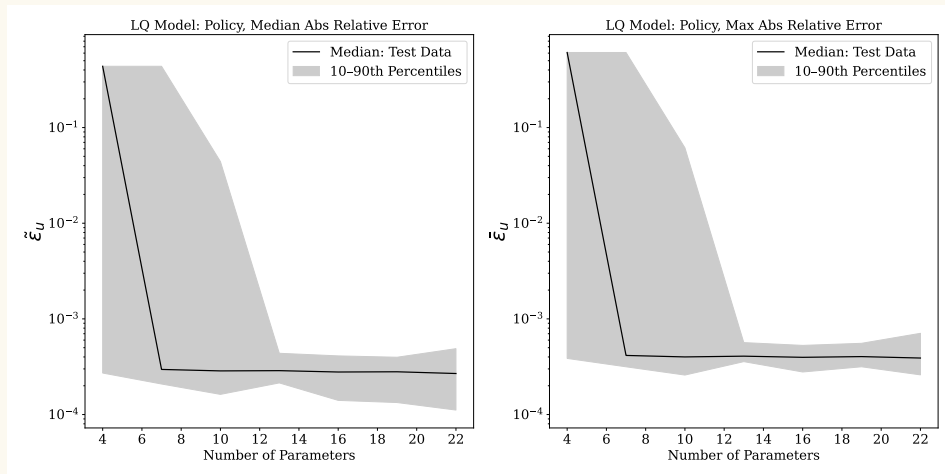
Deep learning is a **non-convex** optimization

- Deep learning is a non-convex optimization.
- In non-convex optimization solutions depends on the initialization of weights and biases θ .
- Most economic dynamical models are non-linear.
- We run the algorithm for 50 different initialization (seed) of the weights and biases.

LQ: train and test Euler-residual loss

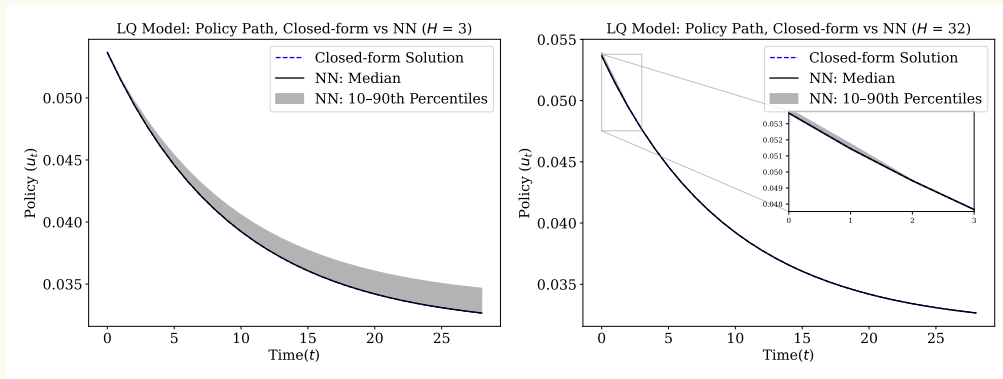


LQ: absolute relative error vs. closed-form solution



- Median (left) and maximum (right) absolute relative error of the NN policy. Both statistics and their cross-seed dispersion fall with network width.

LQ: policy paths, underparameterized vs. overparameterized



- Left: $H = 3$ (underparameterized). Median is roughly accurate, but the across-seed band is wide.
- Right: $H = 32$ (overparameterized). Median tracks the closed-form solution closely and the band collapses.

Application 2: McCall job-search model

From a linear to a non-smooth problem equation

- The LQ model is linear, which makes it the cleanest test but also the easiest.
- Does the blessing survive when the operator $\mathcal{T}$ is nonlinear and the solution is non-smooth?
- The McCall model (McCall, 1970) provides exactly this test: the value function has a **kink** at the reservation wage, so the network must approximate a non-differentiable function.
- An unemployed worker receives i.i.d. wage offers $w \sim f$ on $[0, B]$ each period.
- Accept: permanent income stream $w/(1 - \beta)$. Reject: unemployment compensation c plus continuation value.
- Bellman equation (a nonlinear integral equation with a max operator):

$$v(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta \int_0^B v(w') f(w') dw' \right\}$$

McCall: the kink and why it matters

- Optimal policy: threshold form. There exists a reservation wage $\bar{w}$ such that accept iff $w > \bar{w}$.
- Closed-form value function:

$$v(w) = \begin{cases} \bar{w}/(1 - \beta) & \text{if } w \leq \bar{w}, \\ w/(1 - \beta) & \text{if } w > \bar{w}. \end{cases}$$

- The reservation wage $\bar{w}$ solves:

$$\bar{w} - c = \frac{\beta}{1 - \beta} \int_{\bar{w}}^B (w' - \bar{w}) f(w') dw'$$

LHS: net gain from rejecting the marginal offer. RHS: expected gain from continued search.

- The kink at $\bar{w}$ is a demanding test: the network must learn a non-smooth function from only $N = 12$ training points.

McCall: deep learning implementation

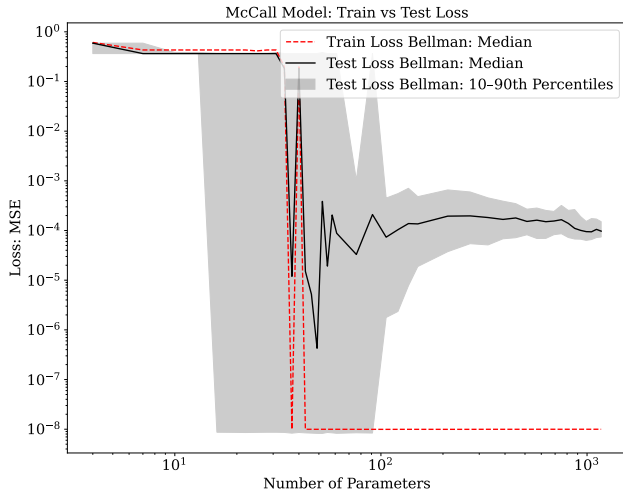
- **Architecture:** one-hidden-layer ReLU network $v(w; \theta_v)$, linear output. 50 random seeds per width.
- **Training:** $N = 12$ uniform grid on $[0, 1]$. Minimize mean squared Bellman residual:

$$\min_{\theta_v} \frac{1}{N} \sum_{w_i \in \mathcal{D}} \left[v(w_i; \theta_v) - \max \left\{ \frac{w_i}{1 - \beta}, c + \beta \int_0^B v(w'; \theta_v) f(w') dw' \right\} \right]^2$$

Integral by quadrature. Adam with early stopping at 10^{-8} .

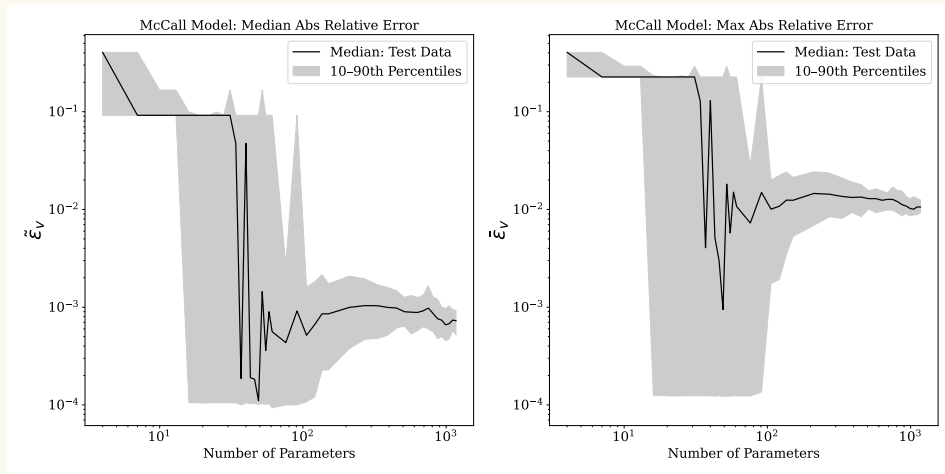
- **Test:** finer $N = 100$ uniform grid, strictly outside the training set.
- **Accuracy metric:** absolute relative error $\varepsilon_v(w) = |v(w; \theta_v^*) - v(w)|/|v(w)|$. Summarized by max and median across test grid, then 10th/50th/90th percentiles across 50 seeds.

McCall: train and test Bellman-residual loss



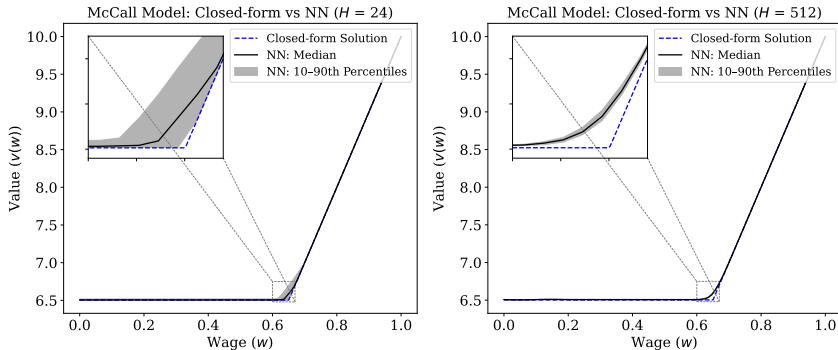
- Test loss descends again in the overparameterized regime: the **double-descent pattern**?
- The across-seed band collapses.

McCall: absolute relative error



- Median (left) and max (right) relative error fall from $\sim 30\%$ to $\sim 2\%$ as the network enters the overparameterized regime.

McCall: value function, underparameterized vs. overparameterized



- Left: $H = 24$ (underparameterized). Median tracks the closed-form solution well, but the across-seed band is wide.
- Right: $H = 512$ (overparameterized). Band is nearly invisible: 50 random initializations converge to the same function.

Application 3: Real Business Cycle model

From deterministic to stochastic, from closed form to none

- The LQ model is linear with a closed-form solution. The McCall model is nonlinear with a kink but still one-dimensional and still admits a closed form.
- The Real Business Cycle model (Kydland and Prescott, 1982) is the most demanding test: a two-dimensional stochastic state space (k_t, z_t) and no analytical solution.
- Social planner, Cobb–Douglas production $e^{z_t} k_t^\alpha$, stochastic TFP:

$$\begin{aligned} \max_{\{c_t, k_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \quad \text{s.t.} \quad c_t + k_{t+1} &= e^{z_t} k_t^\alpha + (1 - \delta)k_t \\ z_t &= \rho z_{t-1} + \sigma \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1) \end{aligned}$$

- Key object: policy function $k'(k, z)$. Benchmark: value function iteration (VFI) on a fine grid.

RBC: the Euler equation

- Euler equation:

$$u'(c_t) = \beta \mathbb{E}_t[u'(c_{t+1}) (\alpha e^{z_{t+1}} k_{t+1}^{\alpha-1} + 1 - \delta)]$$

- **LHS**: marginal utility cost of saving one unit today.
- **RHS**: expected discounted marginal utility gain from extra capital tomorrow (marginal product plus undepreciated capital).
- Nonlinearity plus stochastic two-dimensional state means no analytical solution.
- This is the operator equation from slide 1, now in its full stochastic form.

Parameterization

Description	Symbol	Value	
Discount factor	β	0.96	
Capital share	α	1/3	Cobb–Douglas exponent
Depreciation rate	δ	0.1	per-period capital depreciation
TFP persistence	ρ	0.9	AR(1) coefficient for z_t
TFP volatility	σ	0.01	std. dev. of ε_t
Utility function	$u(c)$	$\ln c$	log utility

RBC: deep learning implementation

- **Architecture:** one-hidden-layer ReLU network $k'(k, z; \theta)$. Output layer: Softplus activation $\log(1 + e^x)$ to enforce $k' > 0$. 50 seeds per width.
- **Training:** 20×20 Cartesian grid ($N = 400$). Euler residual:

$$\ell_{\text{Euler}}(k, z; \theta) = \frac{1}{c(k, z; \theta)} - \beta \sum_{i=1}^M \frac{w_i}{\sqrt{\pi}} \left[\frac{\alpha e^{\rho z + \sqrt{2}\sigma\zeta_i} k'(k, z; \theta)^{\alpha-1} + 1 - \delta}{c(k'(k, z; \theta), \rho z + \sqrt{2}\sigma\zeta_i; \theta)} \right]$$

with $M = 15$ Gauss–Hermite nodes. Adam with StepLR scheduler.

- **Penalty term:** $\lambda(k'(k^*, 0; \theta) - k^*)^2$ with $\lambda = 0.01$, enforcing a necessary implication of the transversality condition at the steady state.

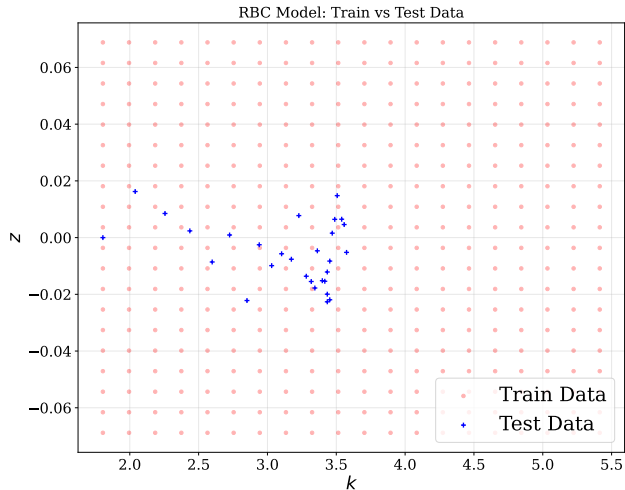
RBC: why the penalty term?

- The Euler equation is a necessary but not sufficient condition for optimality.
- Both the stable and explosive solution manifolds through the steady state satisfy it identically.
- What rules out explosive solutions is the transversality condition (TVC):

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t) k_{t+1} = 0$$

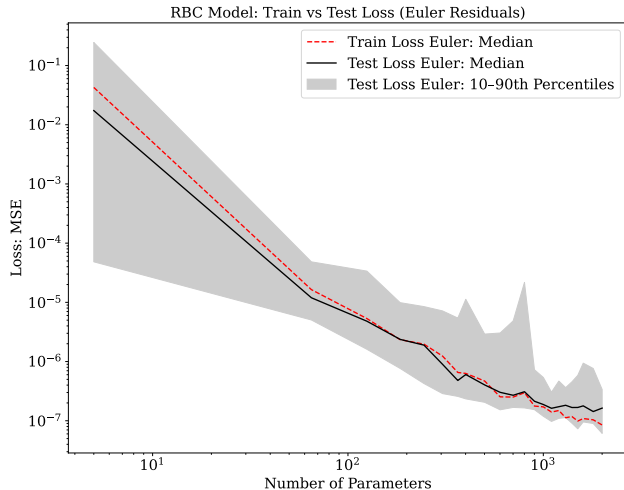
- The Euler residual loss does not enforce the TVC directly.
- The penalty resolves this multiplicity by imposing $k'(k^*, 0) = k^*$: the steady state is a fixed point of the policy function.
- Without this, gradient-based training can converge to the explosive manifold: small Euler residuals but divergent simulated paths.

RBC: training and test data



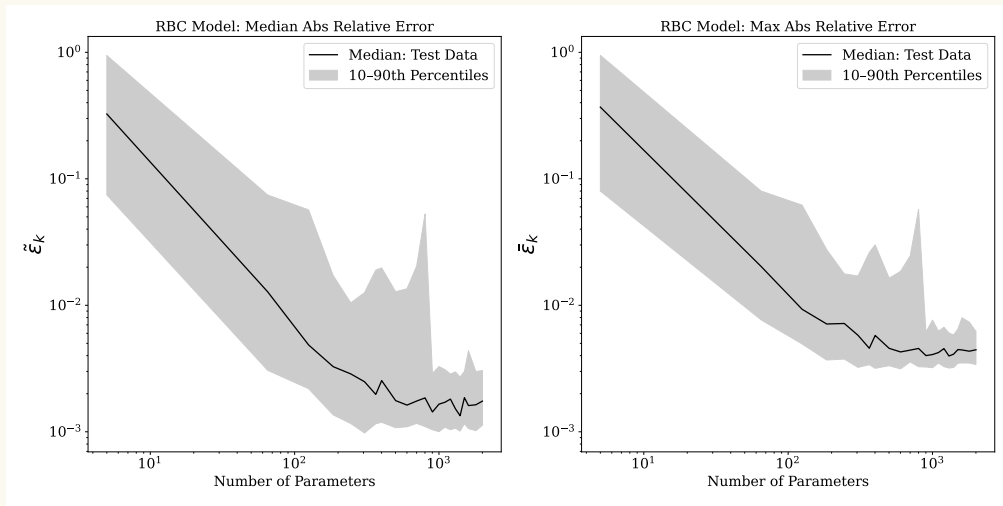
- Red dots: 20×20 training grid.
- Blue crosses: $T = 29$ path simulated under the VFI policy starting at $k_0 = 0.5 k^*$.
- Test path visits the interior of the state space and is never on a training node.

RBC: train and test Euler-residual loss



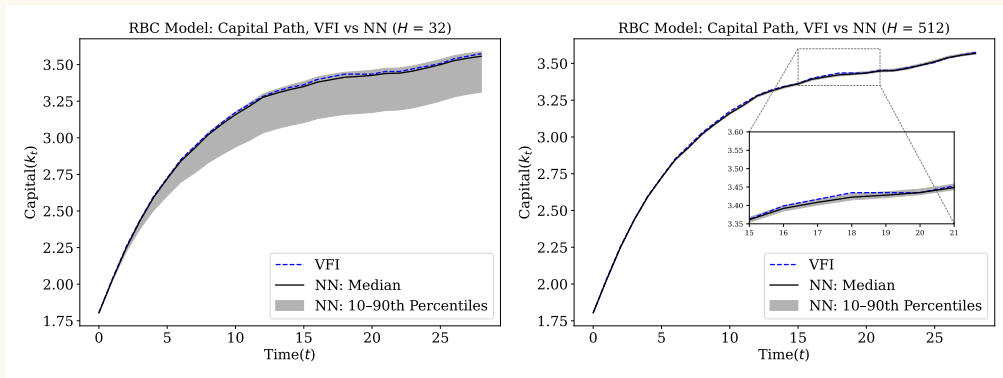
- Both losses decline as the network grows wider, with no **sign of overfitting**.
- The across-seed band narrows sharply: overparameterization simultaneously improves accuracy and reduces initialization sensitivity.

RBC: absolute relative error vs. VFI



- Median (left) and maximum (right) absolute relative error of the NN capital policy versus VFI.

RBC: capital paths, underparameterized vs. overparameterized



- Left: $H = 32$. Median path close to VFI, but across-seed band is wide. Accurate on average but unreliable.
- Right: $H = 512$. Median tracks VFI almost exactly over the full horizon and the band collapses to near zero.

Robustness

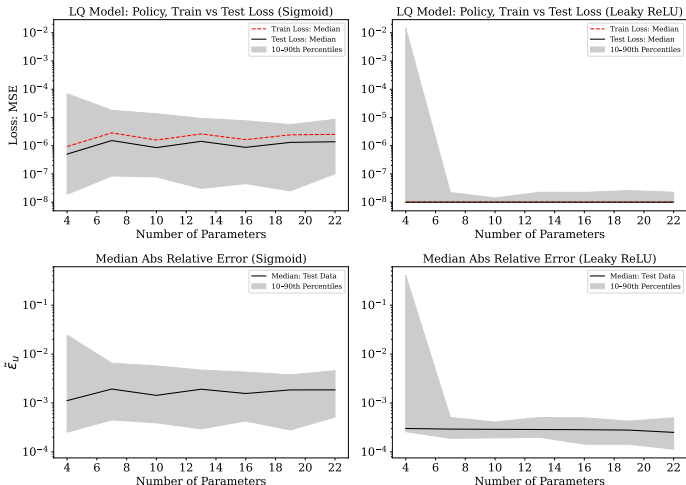
Are these results specific to ReLU and one hidden layer?

- A natural concern: the blessing might depend on the particular activation function or the single-layer architecture.
- We test robustness along two dimensions:
 - **Activation functions:** Sigmoid and Leaky ReLU in addition to the baseline ReLU.
 - **Network depth:** two-hidden-layer architectures in addition to the baseline one-hidden-layer.

Result

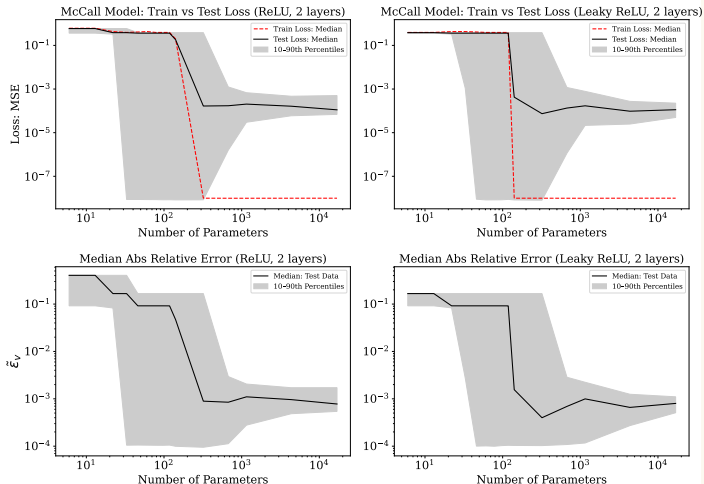
The blessing holds in every configuration we tried. The next slides document this for each model.

Robustness: alternative activations for the LQ model



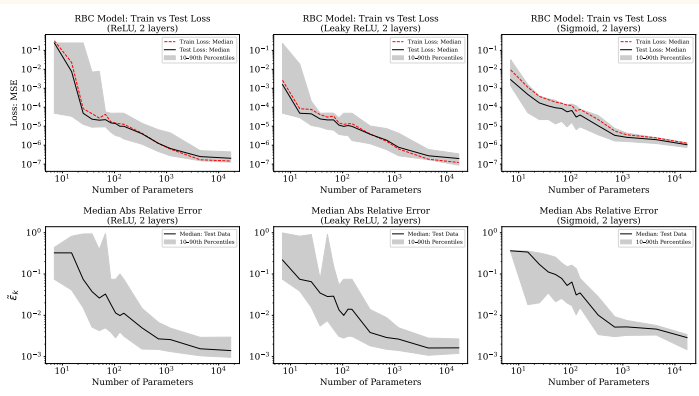
- Sigmoid (left) and Leaky ReLU (right).
- Top row: train/test Euler MSE. Bottom row: median relative error.
- Both activations replicate the main finding: accuracy improves with width, across-seed dispersion collapses.

Robustness: deeper networks for the McCall model



- Two-hidden-layer architecture with ReLU (left) and Leaky ReLU (right).
- Quantitatively similar results: the blessing is not specific to one-hidden-layer networks.

Robustness: deeper networks for the RBC model



- Two-hidden-layer architecture with ReLU (left), Leaky ReLU (center), Sigmoid (right).
- All three activations replicate the main finding for a stochastic model with no closed-form solution.

- The blessing of overparameterization is robust across all dimensions most relevant to applied practice:
 - **Activation functions:** ReLU, Leaky ReLU, Sigmoid.
 - **Network depth:** one and two hidden layers.
 - **Operator type:** Euler residual (LQ, RBC), Bellman residual (McCall).
 - **Solution regularity:** smooth (LQ), non-smooth with kink (McCall), stochastic without closed form (RBC).
 - **State dimensionality:** 1D (McCall), 2D (LQ, RBC).

Conclusion

Takeaways

- **Finding 1:** no overfitting. As network width increases, out-of-sample Euler and Bellman residuals decline. Value and policy functions converge toward their benchmarks.
- **Finding 2:** algorithmic stability. Small networks produce solutions that depend on the random seed. Wide networks concentrate across seeds, making the algorithm reliable and initialization-independent.
- Documented consistently across three operator equations: linear with closed form (LQ), nonlinear with kink (McCall), stochastic with no analytical solution (RBC).
- Robust across activations (ReLU, Leaky ReLU, Sigmoid) and depths (one and two hidden layers).

Bottom line

When in doubt, use a wider network. Overparameterization is a blessing, not a curse.

Appendix

Parameterization: Linear-Quadratic model

Description	Symbol	Value	
Discount factor	β	0.9	
Demand intercept	α_0	1.0	inverse demand level
Demand slope	α_1	1.3	price sensitivity to aggregate
Adjustment cost	γ	100	convex investment cost
Aggregate drift	h_0	0.05	constant in $Y' = h_0 + h_1 Y$
Aggregate persistence	h_1	0.9	AR coefficient for Y

Parameterization: McCall model

Description	Symbol	Value	
Discount factor	β	0.9	
Wage support	B	1.0	upper bound of $U[0, B]$
Unemployment benefit	c	0.1	flow payoff while searching